JUNIOR CERTIFICATE

GEOGRAPHY

Form 2

Study notes

Table of Contents

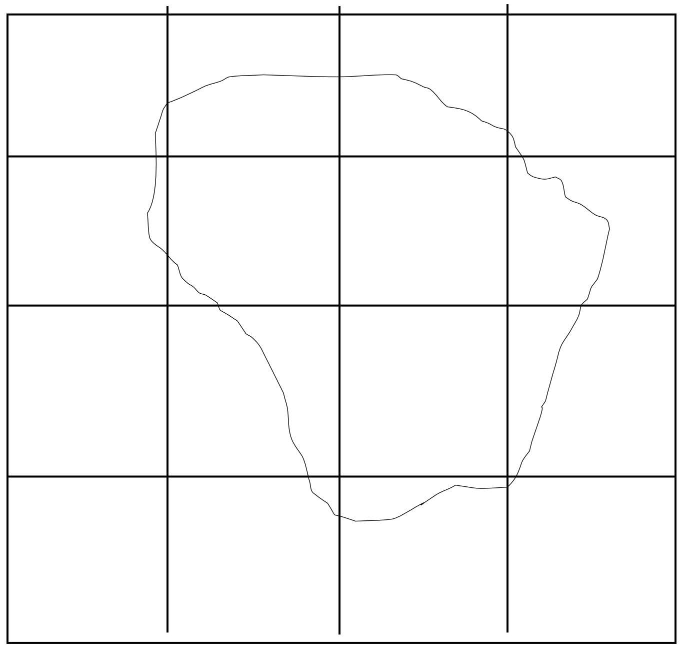
FINDING THE AREA OF AN IRREGULAR SHAPED AREA. 5
WAYS OF SHOWING RELIEF ON THE MAP 5
Hachures 5
Trigonometrical point/station/beacon. 6
Bench mark 6
Spot height 6
Colouring or tinting. 6
Contour lines 7
MAP ENLARGEMENT AND REDUCTION 8
Map enlargement 8
Map reduction 8
Procedure 8
GRADIENT 9
Calculating Vertical Interval (V I) 9
Calculating horizontal equivalence (H E) 10
CROSS-SECTION/ LAND PROFILE. 11
INTERVISIBILITY 12
LATITUDES AND LONGITUDES 13
LATITUDE 13
Uses of latitudes 13
LONGITUDES 14
Some facts about longitudes 14
Uses of longitudes. 14
Using longitudes to tell time 14
The international date line. 15
THE ATMOSPHERE – SEASONS AND CLIMATE 16
SEASONS 16
Winter 16
Spring 16
Summer 17
Autumn (Fall) 17
The cycle of seasons 17

CLIMATE 18
Climatic graphs 18
RELIEF AND DRAINAGE 19
The internal parts of the earth. 19
Causes 19
Lakes in East and Central Africa and their formation 20
The great rift valley: 20
Explanation: 21
Characteristics of rift valley lakes. 21
Volcanic or crater or caldera lakes. 22
Lava dammed lakes 22
Artificial or human made lakes. 23
LAKES IN MALAWI 23
Lake Malawi 23
LAKE CHILWA 23
THE SHIRE RIVER 24
Importance of the Shire River: 24
Cross section of the Shire River. 25
Social economic activities taking place along the Shire River 25
THE STRUCTURE OF THE EARTH 25
Relief regions of malawi 25
The high plateau region (plural plateaux 25
The lower plateau region 26
The rift valley floor 26
RIVERINE FEATURES 27
The upper or youthful stage of a river. 27
Middle river course or the mature stage 27
The lower course or the old stage course. 27
Other riverine features include: 28
FEATURES ASSOCIATED WITH LAKES – COASTAL FEATURES. 29
Peninsulas and capes 29
Headland or point and a bay. 29
Bay 30

Spit 30
Sandbars 30
Shoreline 30
Beach 30
Lagoon 30
Swamps 30
LANDFORMS IN EAST AND CENTRAL AFRICA 31
Highlands or mountains 31
Rivers in east and central Africa. 31
Industrialization 32
Industry 32
Types of industries: 32
- Primary industry 32
- Secondary or the manufacturing industry 32
- Tertiary industry 32
Case study of tourism in Malawi 32
Factors that promote the Malawi tourism industry 32
What Malawi is doing to improve its tourism industry 32
COMMUNICATION, TRANSPORT AND TRADE 33
Types of communication 33
Importance of communication 33
Challenges associated with communication 33
TRANSPORT 33
Types of transport 33
Transport in Malawi 33
Problems of transport in Malawi 34
TRADE 34
Importance of trade 34
Main import commodities of Malawi 34
Malawis main export commodities 34
Balance of trade in Malawi 34

FINDING THE AREA OF AN IRREGULAR SHAPED AREA.

- This is found by adding the total number of full squares to half of the number of shared squares.



Area of full squares $= 2\ km^2$

Area of shared squares $= \frac{12\ km^2}{2}$

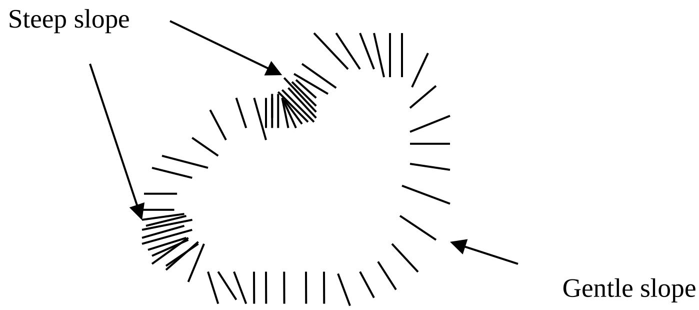
Total area $2 + 6 = 8\ km^2$

WAYS OF SHOWING RELIEF ON THE MAP

- Relief or topography of an area is show by using huchures or colouring, or bench marks or trigonometrical points or spot height or contouring/tinting.

Hachures

- These are short lines drawn around a hill or a mountain. Closely packed lines denote steep slope while sparsely packed lines denote gentle slope. These do not show height above sea level but can show small hills.



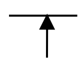
Trigonometrical point/station/beacon.

- These are shown using a small triangle usually with a number to show the height of a peak of a hill or mountain. The number shows the altitude or height above sea level. The height is usually in feet and sometimes in metres) Thus:

 4585 - height

Bench mark

- This is shown by using a horizontal line with an arrow under it. A number beside shows height above sea level.

 4409

Spot height .

Here a spot with a number is used to denote height.

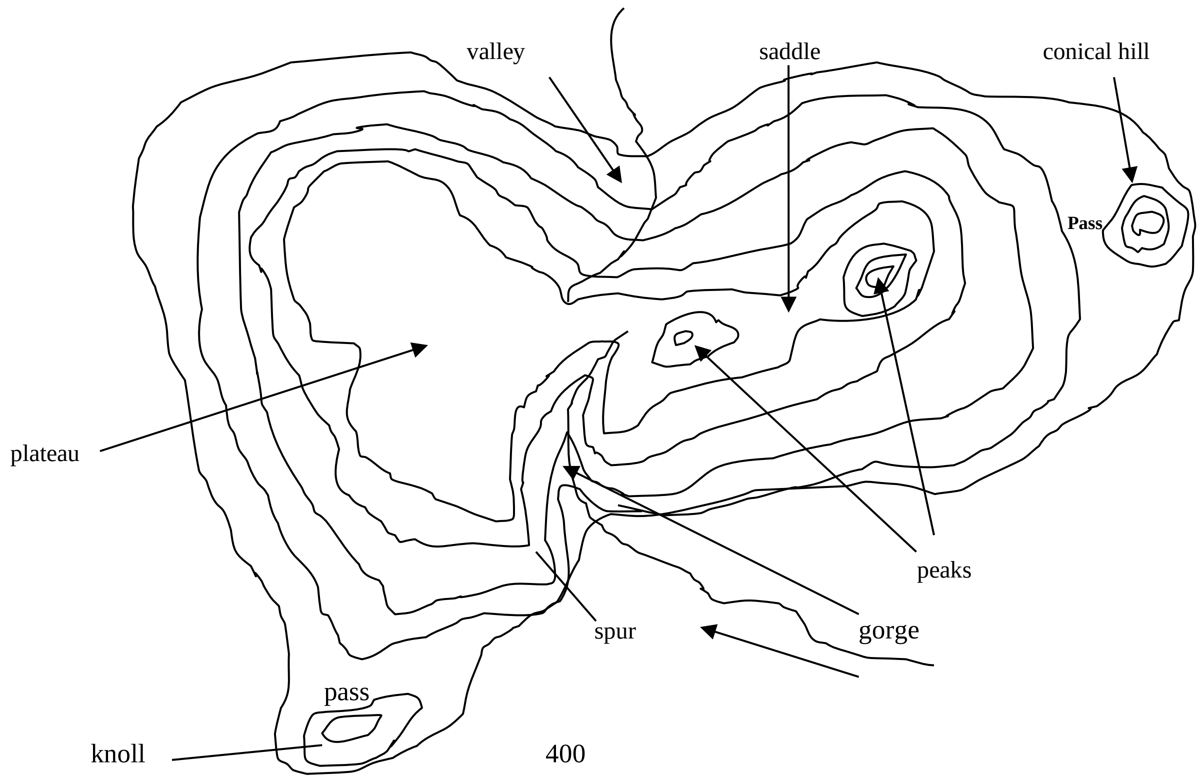
● 1003

Colouring or tinting.

- Green for lowlands and white (snow) for highlands with yellow to brown to purple to pink as intermediate colours

Contour lines

- Contour lines or simply contours are parallel lines drawn to show areas of the same height above sea level.
- The lines are usually spaced 50 feet (approximately 15 metres) apart. Closely spaced contours show steep slope while sparsely spaced ones show gentle or relatively flat area.
- Contours give more details about an area such as valleys, spurs, plateaus, knolls, conical hills, gorges, saddles and passes.



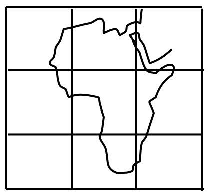
valleys, spurs, plateaus, knolls, conical hills, gorges, saddles and passes.

MAP ENLARGEMENT AND REDUCTION.

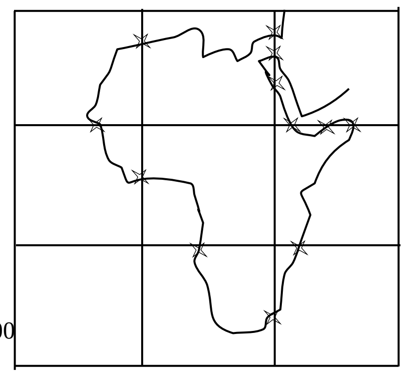
Map enlargement

Map enlargement is done by

- Doubling the distance between the spaces of grids
- Marking all important points on the enlarged map where the required feature intersects the grids on the original map.
- Connect all points using free hand writing. Consider other important angles. **Do not use a ruler.**
- The enlarged map will now have a smaller scale i.e. from 1:50,000 to 1:25,00



original map scale 1:50,000



enlarged map scale 1:25,000

Map reduction

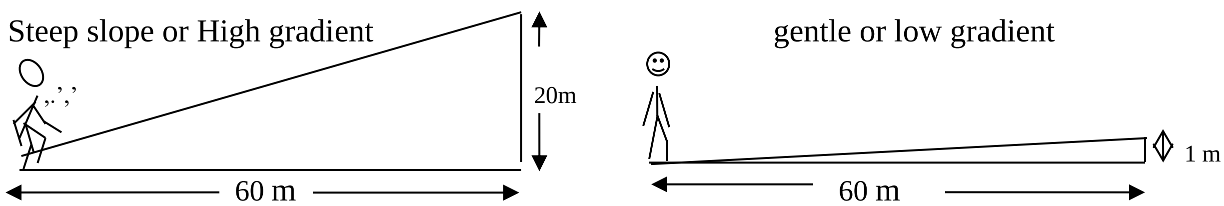
- On map reduction the distance between the grids is reduced by half.
- That is just the opposite of map enlargement.
- Thus, the spaces are now reduced from say two centimeters apart to one centimeter apart.

Procedure

- Draw grids one cm apart (half the original distance between grids).
- Mark all important points on the reduced map where the required feature intersects the grids in relation to the original map.
- Connect all points using free hand writing (do **not** use a ruler unless there is a perfect straight line).
- The scale of the map is now enlarged or doubled. That is from 1:50,000 to 1:100,000.

GRADIENT

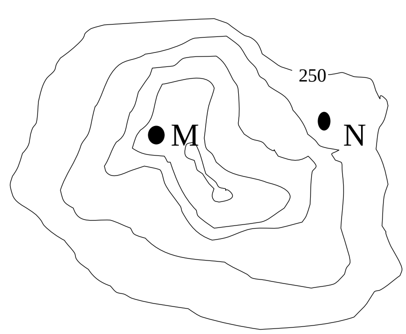
- Gradient refers to the rise or fall a person gains after at a given distance. It is also the sharpness of a slope. An area of high gradient has a steep slope while an area of low gradient is relatively flat.



- Gradient is calculated by dividing the vertical interval (height gained of lost between the lowest contour line and the highest one) by the horizontal equivalence (distance covered). Thus:

$$\text{Gradient} = \frac{\text{vertical interval}}{\text{Horizontal equivalence}} \quad \text{or} \quad G = \frac{V\,I}{H\,E}$$

- The answer is usually expressed as a ratio such as 1:70 meaning there is a rise or fall of 1 metre for every 70 metres horizontal distance travelled. The numerator should **always** be reduced to 1.



Calculating Vertical Interval (V I)

- Identify the value of the contour lower than the lower point (N) and value of the contour lower than the higher points (M) in question.

- Subtract the value of the lower contour (300) from that of the higher one (400).
- If the values of the contours in question are not given, the other higher or lower contours can help you to establish the values of the contours in question since the difference from one contour is usually 50 feet (approximately 15 metres).
- Thus each time you cross a contour upwards or downwards, you add or subtract 50 feet (15 metres) respectively.
- The answer you get is the vertical interval.

Remember to convert your answer into metres by dividing it by 50 then multiply the answer by 15.

Thus: $400-300=\frac{100}{50} \times 15 = 30$

Our vertical interval is 30m

Calculating horizontal equivalence (H E)

- This is calculated in the same way distances are calculated on maps. **Take note that the answer you get is usually in kilometers, hence, it should be converted to metres by multiplying it by 1000**.
- After getting the both the vertical interval and the horizontal equivalence, divide the vertical interval by the horizontal equivalence. Thus

$\frac{VI}{HE}$ e.g $\frac{30}{1500}$ the answer is always simplified to $\frac{1}{X}$ or 1:X

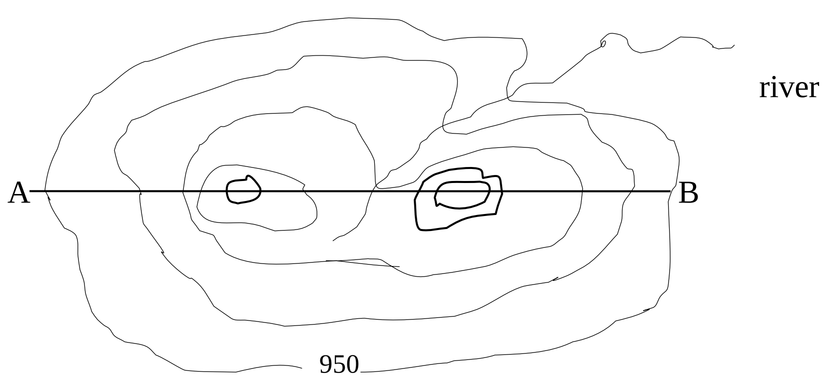
- In our example above the final answer becomes 1:50. If a fraction is left, bring it to the nearest whole number.
- Gradient is important in road and rail construction.
- A gradient of 1:60 makes an angle of 1^0 to the horizontal.
- Railways need to be constructed with gradient not less than 1:60. (1^0 angle) otherwise trains will have difficulties in moving on such gradient.
- Cars can climb an angle of 30^0 while bicycles cannot.

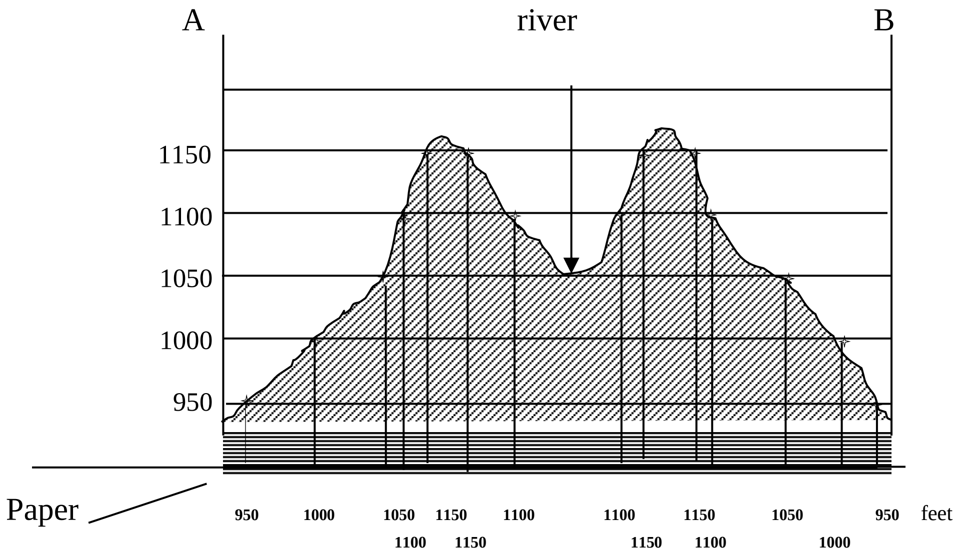
CROSS-SECTION/ LAND PROFILE.

- This refers to the side view of an area between two points.
- It includes the highlands and valleys along the two points.
- It is drawn using horizontal lines **equally spaced (same vertical interval)**.
- The vertical interval is usually 1cm representing 50 feet (interval between contours) on the map.
- The vertical interval between the horizontal lines is also known as the **vertical exaggeration** just because the cross section to be drawn may **not give the exact shape** of the area.
- It will show where highlands or lowlands are but somehow exaggarated.
- Two perpendicular lines are drawn and labeled one at each end of the horizontal lines to mark the two points in question.

Make sure that

- The distance between the two points on the horizontal line is equidistant to same points on the map and that spaces between the horizontal lies are exactly 1 cm apart.
- The lines connecting heights should be drawn with **free hand** and that this line at top most part of the mountains should **curve upwards** and **downwards** where there is a valley or river.
- The part below the line of cross-section should be **shaded**.





INTERVISIBILITY

- This refers to being able to see or not see certain point from a given point and vice versa.
- Intervisibility is there **only when there is no barrier** such as a high land mass in between the two points.
- There is intervisibility if a straight line drawn connecting the two points in question does not cross a barrier (an area higher than it). If this line crosses a barrier then there is no intervisibilty.
- In the cross-section above, there is intervisibility between the first and second mountain peaks because there is no barrier in between. On the other hand there is no intervisibility between point A and the river or the second peak because there is a barrier (the first mountain peak) in between them.

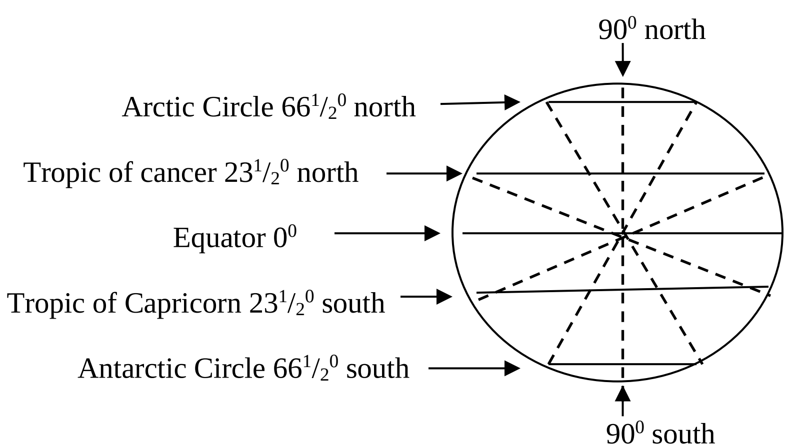
LATITUDES AND LONGITUDES

LATITUDE

- A latitude is an imaginary line measured in degrees north or south of the equator.
- They are also known as **tropics** or **parallels**.
- There are about 111km in every 1^0 interval.
- The **equator** is the chief latitude. It is marked $\mathbf{0^0}$. It also divides the world into the **Northern** and **Southern Hemispheres**.
- In Africa it passes through countries like Gabon, Congo Brazzaville, Democratic Republic of Congo, Uganda, Kenya and Somalia.

The other main latitudes are:

- Tropic of cancer – marked $23^1/_2{}^0$ north
- Tropic of Capricorn – marked $23^1/_2{}^0$ south
- Arctic Circle – marked $66^1/_2{}^0$ north
- Antarctic Circle – marked $66^1/_2{}^0$ south.
- Latitudes between the equator and latitude 30^0 north or south are known as low latitudes.
- Those between 30^0 and 60^0 north or south are known as mid latitudes.
- Those between 60^0 and 90^0 north or south are called high latitudes.



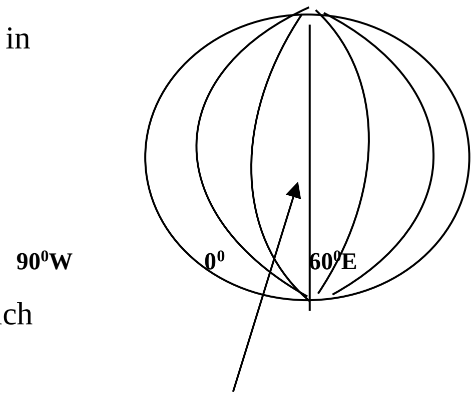
Uses of latitudes

- To locate places on a map e.g. Malawi is located between 9^0 and 18^0 south
- To determine climate of a place–
- Places around low latitudes have higher temperatures than mid or high latitudes
- Places around equatorial latitude have high rainfall than the others.
- Areas around latitude 30^0 north or south receive the least rainfall

LONGITUDES

- A longitude is an imaginary vertical line measured in degrees east or west of the prime meridian.
- They are also known as meridians.
- **The Prime Meridian** is the chief longitude.
- It is also known as the **Greenwich**
- **Meridian** because it passes a place called Greenwich east of London. It is marked 0^0.

  All longitudes east or west start counting from this meridian.



The Prime Meridian

Some facts about longitudes

- All longitudes are equal in length.
- They all meet or intersect at the North or South Pole.
- All latitudes together with their corresponding ones on the other side of the earth form what we call great circles.
- Every great circle divides the earth into two equal hemispheres
- The longitude marked 180^0 is the continuation of the Prime Meridian.
- In every 15^0 there is a difference of on hour, thus 4 minutes in every one degree.

Uses of longitudes.

- They are used to tell local time.
- They are used to locate places on a map, e.g Lilongwe is 33^0 east or 33^0 E

Using longitudes to tell time

- Places east of the Prime Meridian are ahead of time and those west of it are behind time.
- As such, one hour is added in every 15^0 eastwards and one hour is subtracted every 15^0 westwards. Besides, for every 1^0, there are 4 minutes. As such, 73^0 becomes 73 x 4 minutes = 292 minutes.
- Divide this by 60 to get hours. We get 4 hours and 52 minutes.
- We add this time when going east or subtract this time when going west.
- All places along or close to the same longitude are likely to have the same local time.

- For example Blantyre, Balaka, Durban and Cairo are along or close to the same longitude, (30^0 E), hence, uses the same local time.
- In some cases to avoid political and economical confusions some countries may agree which longitude to use when determining local time.
- For example Namibia uses the 30^0 time zone despite belonging to the 15^0 time zone. China uses the Beijing time despite having some areas almost five hours away.

The international date line.

- This is an imaginary line along the 180^0 longitude which when one crosses it he or she gains or loses a day.
- When one crosses it to the west he or she gains a day.
- On the other hand when one closes it to the east he or she loses a day.
- That is to say when it is Monday on the west it is still Sunday on the east of it.
- However the International dateline does not always follow the 180^0 longitude.
- It zigzags in some places to avoid cutting some islands and land masses along the 180^0 longitude.
- Such places include Fiji and Tonga islands and the land mass eastern Asia.
- For political and economical purposes other countries just chose which side to belong to despite being geographically on the other side of it.

THE ATMOSPHERE – SEASONS AND CLIMATE

SEASONS

- A season is the division of a year associated with atmospheric conditions characterized by intensity of radiation, precipitation and wind direction.
- The major seasons include winter, spring, summer and autumn.
- Malawi has only three main seasons: warm-**wet season** (summer) – November to March, **cool dry season** (winter) – April to July and **hot dry season** (spring) – August to October.
- Generally, there are four seasons namely, winter, spring, summer, and autumn (fall).

Seasons in the Southern Hemisphere

Winter

- Occurs from June to August

Characteristics

- Low temperatures.
- Low precipitation
- Short days and long nights.
- Winter solstice 21 June when the overhead sun is at the furthest in the northern hemisphere.
- Continuous nights at the South Pole

Spring

- Occurs from September to November

Characteristics

- Temperatures increasing
- Days becoming longer
- Dry and hot
- Equinox on 21 Sep (over head sun at the equator as it crosses from the N Hemisphere to the S. Hemishere , hence, equal length of day and night.

Summer

- Takes place between December and February.

Characteristics

- High temperatures.
- High precipitation.
- Longer days than nights.
- Sun overhead in the S Hemisphere
- Summer solstice on 22 December (the sun is overhead at Tropic of Capricorn).
- Continuous sun shine at the a south pole

Autumn (Fall)

- Experienced between March and May.

Characteristics

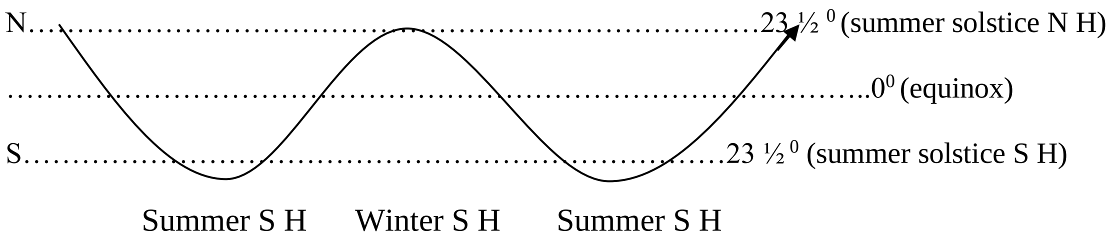
- Temperatures getting cooler.
- Precipitation getting low.
- Equinox on 21 March (the sun is crossing the equator to the N Hemisphere
- NB autumn is not well developed in the S Hemisphere due to too much ocean surface which influence much of the weather patterns over the land.

The cycle of seasons

- The cycle of seasons explains the inversion of seasons from one hemisphere to another. The table below explains this scenario.)

MONTHS	S HEMISPHERE	N HEMISPHERE
June to August	Winter	Summer
September to November	Spring	Autumn
December to February	Summer	Winter
March to May.	Autumn (Fall)	Spring

The revolution of the earth and apparent movement of the sun across the sky.

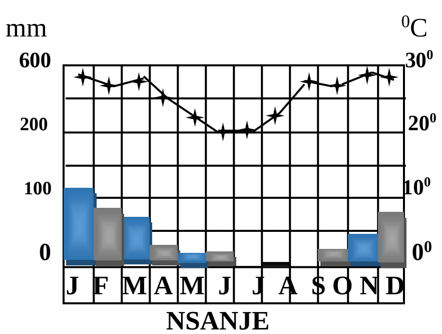


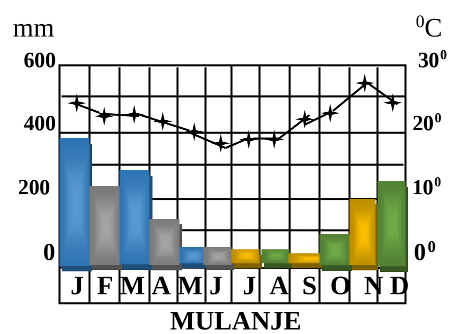
CLIMATE

- Climate refers to atmospheric conditions of a particular place over long period of time.
- Malawi's climate is known as tropical continental or savanna climate.
- However, there are some climatic variations within Malawi from place to place.
- Lowlands like Nsanje receive little rainfall
- Highlands like Mulanje receive high rainfall due to the highness of the area facing rain bearing trade winds from the Indian Ocean.
- Places like Nkhata Bay and north of Karonga receive high rainfall because the area faces the rain bearing trade winds (South-east on-shore winds from Lake Malawi).
- Lilongwe receives moderate rainfall due to its plateau position.

Climatic graphs

Climatic data for some areas in Malawi





RELIEF AND DRAINAGE

- Relief refers to the physical shape of an area such as altitude, landforms and slopes. Others refers it to topography especially when it includes artificial features.
- Examples of relief features include valleys, mountains, hills, plains, valleys, lakes, and rivers. But how were these features formed?
- To understand this, we need to understand the internal structure of the earth.

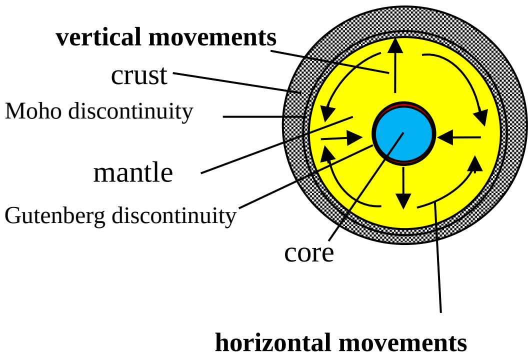
The internal parts of the earth.

The earth has three main layers. Thus:

- The crust or lithosphere: a solid top most layer which forms continents and ocean floor. It has heavier material on ocean floors then on continents.
- The mantle or mesosphere: this is a layer of a hot molten (liquid or plastic) below the crust. It makes about 80% of the earth's mass.
- The molten material is in continuous motion.
- This motion causes the crust to break and move to and fro leading to the formation of relief features on the earth's surface.
- The core: this is the innermost layer of the earth.
- The boundary between the core and the mantle is called the Gutenberg discontuity while the one between the crust and the mantle is called the Mohorovicic or Moho discontinuity

Causes

Vertical (epierogenic) and horizontal (lateral /orogenic) movements

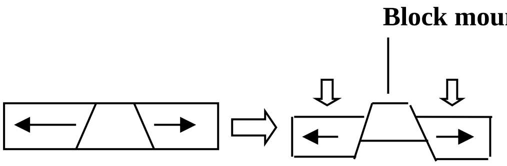


[a] *Vertical movements* -These run between the core and the crust causing the later [crust] to break .

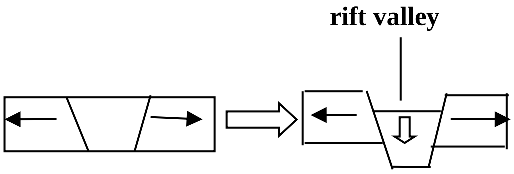
[b] *Horizontal movements* -

- These run across plates causing the crust to fold or break due to compressional or tension forces .
- The movements inside the earth's internal parts influence tensional or compressional forces on the crust which causes faulting and folding, hence, giving birth to relief features like

mountains, rift valleys, lakes and plains. Mountains formed by faulting are known as block mountains



tensional force



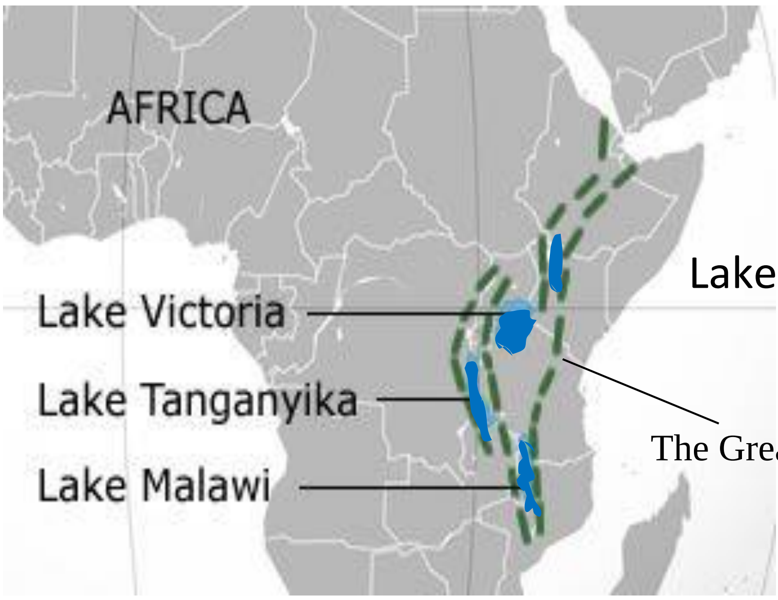
compressional force

Lakes in East and Central Africa and their formation

- A lake is a large hollow in the earth's surface filled with water.
- Basing on their formation lakes can be grouped as follows:
- Lakes formed by the earth's internal movements sub-grouped into rift valley lakes, basin lakes, crater lakes and lava dammed lakes.

The great rift valley:

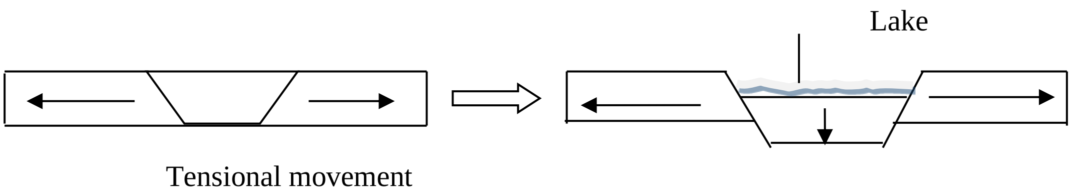
- The Great Rift Valley starts from the area along Jordan River, through the Sea of Galilee, the Red Sea, Lake Turkana, then forms two arms -the west arm and the east one with Lake Kivu and Tanganyika on the west arm.
- Then the two arms meet just north of Lake Malawi and then the rift valley passes along lake Lake Malawi, Lake Malombe and the Shire River valley.

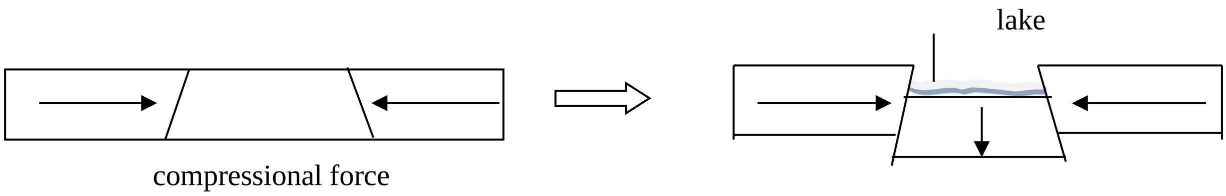


Lake Turkana

The Great Rift Valley

- **Rift valley lakes***:* Formed by faulting of the crust.





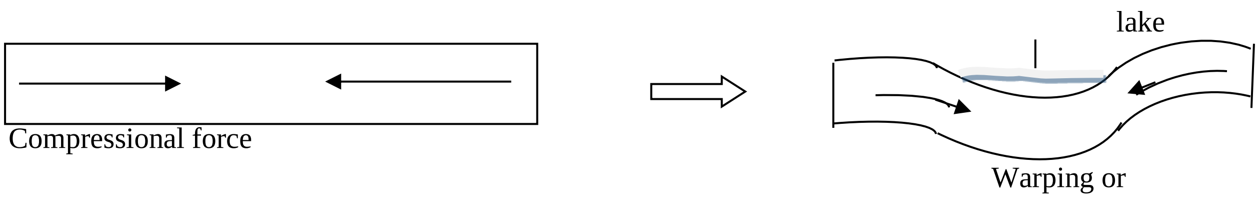
Explanation:

- The earth's internal movements causes faulting and tensional or comressional movements.
- The tensional or compressional forces make the central crustal block to form a lift valley or a hollow. When the rift valley or hollow is filled with water, a rift valley lake is formed.
- When the rift

Characteristics of rift valley lakes.

- Rift valley lakes are usually long, narrow, deep, seem to be linear, may have steep sides are found along crustal faultlines or plate boundaries. Being deep these lakes are more permanent than the other lakes.
- Examples of rift valley lakes include Lake Turkana, Lake Kivu, Lake Albert, Lake Edward, Lake Tanganyika and Lake Malawi.

- *Basin lakes:* Formed be warping or folding of the crust



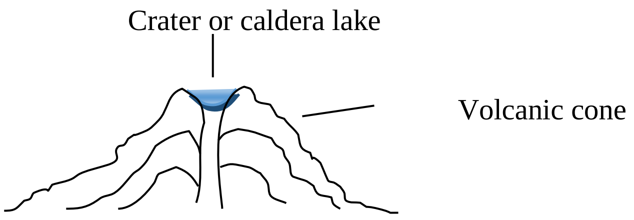
Folding

- Basin lakes are usually round and shallow.

- Examples include Lake Victoria, Lake Chad and Lake Chilwa. these lakes are usually round and are shallow.
- Such lakes are more temporary than rift valley lakes.

Volcanic or crater or caldera lakes.

- These are formed when volcanic hollow called crater caldera is filled with water.
- A crater is cup-like or bowl-like feature formed at the mouth of a volcanic cone.
- Most of these lakes are very small and more temporary such that most of them are extinct.

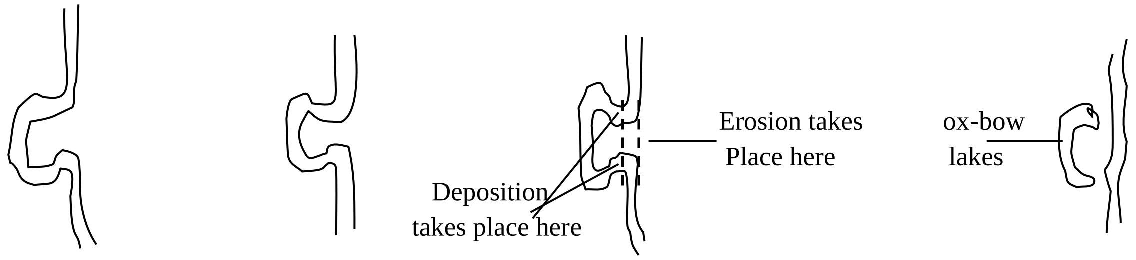


Lava dammed lakes.

- These are farmed when lava flow blocks a river valley to form a lake upstream.
- Examples include Lake Kineret (Lake Tiberias or Sea of Galilee) – not in Africa.

(v). Ox bow lakes

- These are formed when a river bend or meander is cut off.
- This usually occurs in the lower course of a river where the river is at its old age stage.
- Here the river flows so slowly that it ends up depositing its load into its own channel. With lateral soil erosion active at this course, the river changes its direction by cutting a new channel.
- The entry and the exit of the old channel are then filled deposits of sediments to form what we call an ox-bow lake.
- There are many ox-bow lakes along the lower Shire Valley.



Artificial or human made lakes.

- These are formed by damming rivers. Such lakes include lake Nasser (Aswan High Dam) on the Nile River in Egypt, Lake Kariba along the Zambezi River along Zambia and Zimbabwe border, Lake Cabora Bassa on the Zambezi River in Mozambique and Lake Volta along Volta River in Ghana.

LAKES IN MALAWI

Lake Malawi

- Latitude: 9^0 28^0 – 14^0 12^0 S. Longitude: 34^0 E
- Length: between 550 and 580 kilometres, width 20km off Senga Bay to over 80km at Chintheche (about 365 miles and about 52 miles wide, hence also known as the calendar lake).
- Area: approximately 24,000km^2 – about one fifth of Malawi's area. It is the third largest in Africa after Lake Victoria and Lake Tanganyika.
- Depth: 704 metres near Usisya (232 metres below sea level). It is the second deepest in africa after Lake Tanganyika.
- Main tributaries: Songwe, Lufira, North Rukuru, South Rukuru, Lweya, Dwangwa, Bua, Linthipe, and Bwanje in Malawi. Domwe and Ruhuhu in Tanzania.
- Outlet: Shire River.
- Main island: Likoma, Chizumulu, Mbenje, Domwe, and Maleri.
- Main peninsulas: Nankhumba or Cape Maclear and Luromo.
- Main swamps or mashes: Lweya-Kawiya, Limphasa, Bana north of Dwangwa river and Dzadza at Bua river points.
- Main logoon: Chia lagoon in Nkhotakota

LAKE CHILWA

- Type: inland drainage (basin) lake.
- Length: about 40 km long
- Width: about 27 km
- Depth: 3-5 m only.
- Main tributaries: Phalombe and Sombani from Mulanje Mounyain, L ikangala and Domasi from Shire Highlands (Zomba Plateau) and Mnembo from Mozambique.

- Outlet: no outlet/inland drainage, hence, salty.
- Main islands: Chisi and Thongwe.

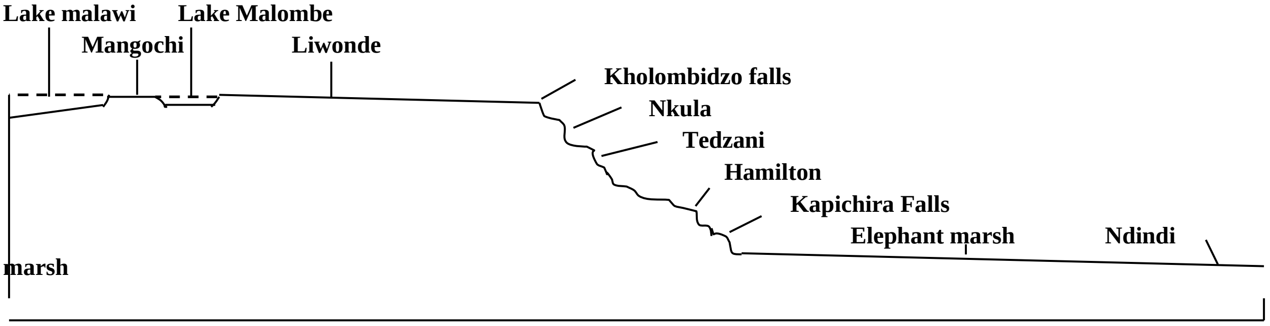
THE SHIRE RIVER

- Length about 400 km on the Malawi part.
- Courses or sections:
  - Upper part which is flat with Lake Malombe dropping about 5cm/km.
  - There is a barrage at Liwonde which controls the flow of the river to form a water reservoir upstream.
  - The upper course almost flat –having lower river course characteristics.
  - Howe there are almost no ox-bow lakes here due to little or no deposition taking place at this section.
  - Middle part with youthful stage characteristics such as water falls and rapids.
  - It is about 80 km long. It starts from Matope to Kapichira falls.
  - Drops about 485/km. It is in this section where there are waterfalls such as Kholombizo, Nkula, Tedzani, Hamilton and Kapichira as well as Mpatamanga Gorge are found.
  - Lower part which is gentle but faster than than the upper course there are a number of oxbow lakes such as Lisuli near Chikwawa boma.
  - Here, the river drops about 40cm/km.
- Main marshes: Elephant and Ndindi.
- Main tributaries: Rivirivi, Lisungwe, Wamkulumadzi, Mwanza from Kirk Range and Ruo from Mulanje Mountain.
- Main waterfalls: Kholombozo, Nkula, Tedzani, Hamilton and Kapichira.

Importance of the Shire River:

- For irrigation at Nchalo Sugar Estate.
- For fishing along the river itself and at Elephant and Ndindi marshes.
- For hydro electricity production at Nkula, Tedzani and Kapichira.
- For water supply to Blantyre city. The water is tapped at Walkers' Ferry.

Cross section of the Shire River.



Social economic activities taking place along the Shire River

Some of them include:

- Fishing
- Water supply to Blantyre city residences.
- Production of hydro-electricity.
- Irrigation farming in the lower shire including Nchalo irrigation scheme
- Agricultural activities

THE STRUCTURE OF THE EARTH

Relief regions of malawi

Malawi has three main relief regions namely

1. The high plateau region
2. The lower plateau region
3. The rift valley floor region.

The high plateau region (plural plateaux)

This comprises of Malawi's main mountains such as

- Nyika, Viphya, Misuku and Mafinga in the northern region;
- Dowa, Dedza, Dzalanyama and Kirk Range in central region;
- Zomba, Kirk Range and Mulanje in the south.

The lower plateau region

- This include plain areas such as Mzimba, Kasitu and region
- Lufira plains in the north; much of the plain areas in central Malawi which includes Lilongwe – Kasungu plain;
- Shire Highlands Lake Chilwa- Phalombe plains in the south. Namwera hills lie to the north of lake Chilwa-Phalombe plain while Mulanje is at the south, Zomba is at its west. Most of Lake Chilwa basin is a swamp.
- **Note** that these lower plateaux regions have some region isolated mountains or hills (island-like mountains within them which are known as **inselbergs**. Such inselbergs include
- Hora Mountain in Mzimba plain,
- Ngala, and Bunda hills in Lilongwe Kasungu plain,
- Mulanje and Chiradzulu in Shire Highlands, high plateau
- Mpyupyu in Lake Chilwa-Phalombe plain
- Some parts of plains where some rivers join the lake like parts of Lake Chilwa-Phalombe plain are so flat that there are no slopes for tens of kilometers due to prolonged deposition.
- Such plains are known as **lacustrine** plains. Such places are good for agricultural production.

The rift valley floor.

This in Malawi is divided into two parts

- The northern part, occupied by lake Malawi and its lakeshore areas. It is wider at the far north but narrow with steep sides between Chitimba and Nkhata Bay.
  - This part of the rift valley lies between Nyika and Viphya on the malawi side and Livingstone Mountains on the Tanzanian side from here the valley widens as we move southwards.
  - At the tip of the lake, Chilipa hills which extends to form Nankhumba or Cape Maclear peninsula divides the rift valley into the western and eastern arms
- The Shire Valley to the south. This extends from Lake Malawi to the Lower Shire.

RIVERINE FEATURES

- Riverine feature are the things found along a river bank and usually formed by the river itself.
- Examples include gorges, waterfalls, watershed, catchment area, confluences, levees, meanders, oxbow lakes, floodplains and deltas.
- A river course is a journey the river makes from its source to the sea.
- Some of these features are commonly found on the on the **upper course**, some in the middle and some in the **lower course or part** of the river.

The upper or youthful stage of a river.

- This upper part of the river is usually referred to the **youthful (juvenile)** stage of the river just because the river is usually energetic or flows fast just like young animal.
- This is a **zone of erosion** (mostly vertical erosion). Hence, most of the features here are products of erosion. Such features include:
- **Gorge:** A narrow and steep V shaped valley through which a river flows. e.g. Mpatamanga gorge along the Shire River.
- **Waterfall or fall:** this is a descent of water when a river passes across a hard rock to a soft rock downstream.
- The lower soft rock is eroded faster hence resulting in a sudden steepness of the river course. (The Shire River has its upper course features in its middle course)

Middle river course or the mature stage.

Some of the features or characteristics found include

- The river gets wider
- The land on which the river flows start to become flatter.
- Sand bars and islands due to deposition of stones and gravel because the river current no longer has the force to carry them further.
- River braiding (dividing into a number of channels) disappearing.
- A lot of tributaries join the river.
- Some meanders
- This is also a zone of lateral/horizontal erosion.

The lower course or the old stage course.

- This is This is also known as the **old age stage** just because of the gentle slope or gradient which makes the river flow slowly like an old person.

- The slowness of the river makes it off load or deposit its **load** or **materia**l within its own channel, hence making this course also known as the **zone of deposition**.
- Common features include:
  - Widest river.
  - **Meanders:** river bends or winds. Lateral (side wards) erosion makes the river wind avoiding the load deposited along its channel resulting into meanders.
  - **Oxbow lakes :** these are small lakes formed when meanders are cut off.
  - **Levees:** these are raised river banks formed due to deposition of load during floods. The friction of the water and the sides of the river makes the side waters flow slowly hence encouraging deposition on the sides of the river forming raised banks called levees.
  - **Floodplains:** these are relatively level part of the river valley resulting from alluvium deposition of a river during floods. This is characterized by river braiding (dividing into several channels), levees and oxbow lakes.
  - **Delta:** This is a fan shaped area formed by alluvial deposition at the mouth of a river usually making the river form many **distributaries** (smaller branches).

Other riverine features include:

- **Confluence:** this is a place where two rivers meet. The smaller one is called a **tributary.**
- **Catcment area.** This is the river basin or drainage area. It is also the exposed part of the **acquifer** which receives rainfall. The border of this river basin is called the **watershed.**
- **Watershed**: This is a water parting area separating two river systems. It is also an elevated area which separates two river systems or two continuous drainage areas.

River profile/ cross section

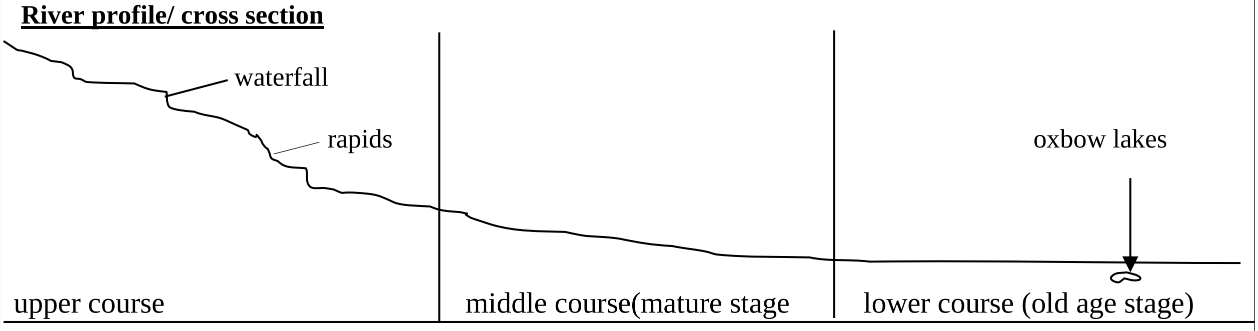
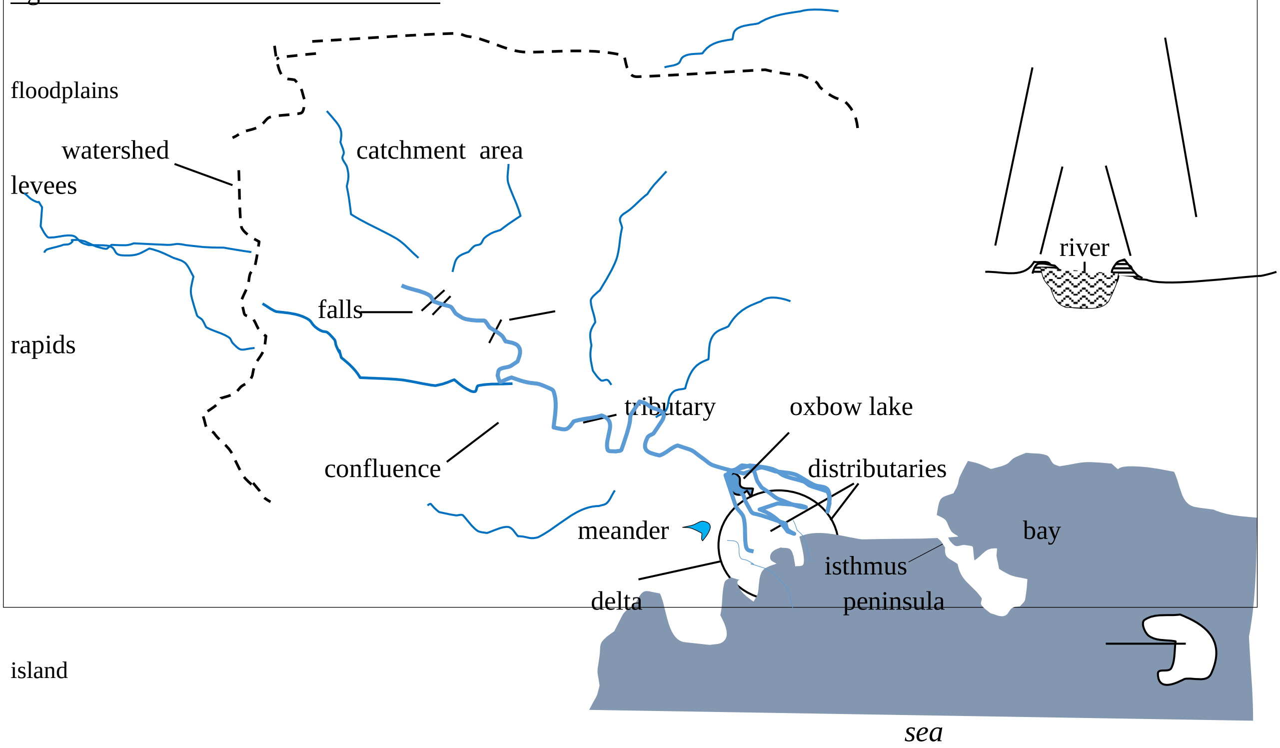


fig. riverine and some shoreline features



FEATURES ASSOCIATED WITH LAKES – COASTAL FEATURES.

- These are features that are commonly found around or near the lake.
- These include peninsulas or capes, headlands and points, bays, beaches, spits, sandbars, logons, swamps, deltas and estuary.

Peninsulas and capes

- A peninsula is a finger of land mostly surrounded by water by still attached to the mainland by a narrow neck. The narrow neck is called an isthmus.
- Luromo and Namkhumba or Cape Maclear are examples of peninsulas around Lake Malawi.
- When a peninsula is found at the tip of continents it is usually called a **cape**

Headland or point and a bay.

- A headland or a point is a land which sticks out into the lake.
- Examples of points or headlands on Lake Malawi include Lion point which together with Luromo peninsula forms Young’s Bay.
- Other points include, Chiromo, Msuli, Bandawe, Bua, Kachulu and Kambiri points.

Bay

- The part of the water that curves into the land between two points or headlands is called a **bay**.
- In other words a bay is just the opposite of a headland or a point.
- Examples of bays on Lake Malawi include Wissman, Young, Nkhata Bay, Domira, Senga, Monkey and Nkhudzi bays.

Spit

- This is a low line of sand of sand running from the land across a bay.
- It is also a narrow strip o land stretching into the lake. They are sand deposits built by long shore currents.

Sandbars

- A sandbar is a low hill of sand formed deposition.
- There is a sand bar separating Lake Malawi and Lake Malombe and another between Lake Chilwa and Lake Chiuta.

Shoreline

- This is a point where the lake meets the land.

Beach

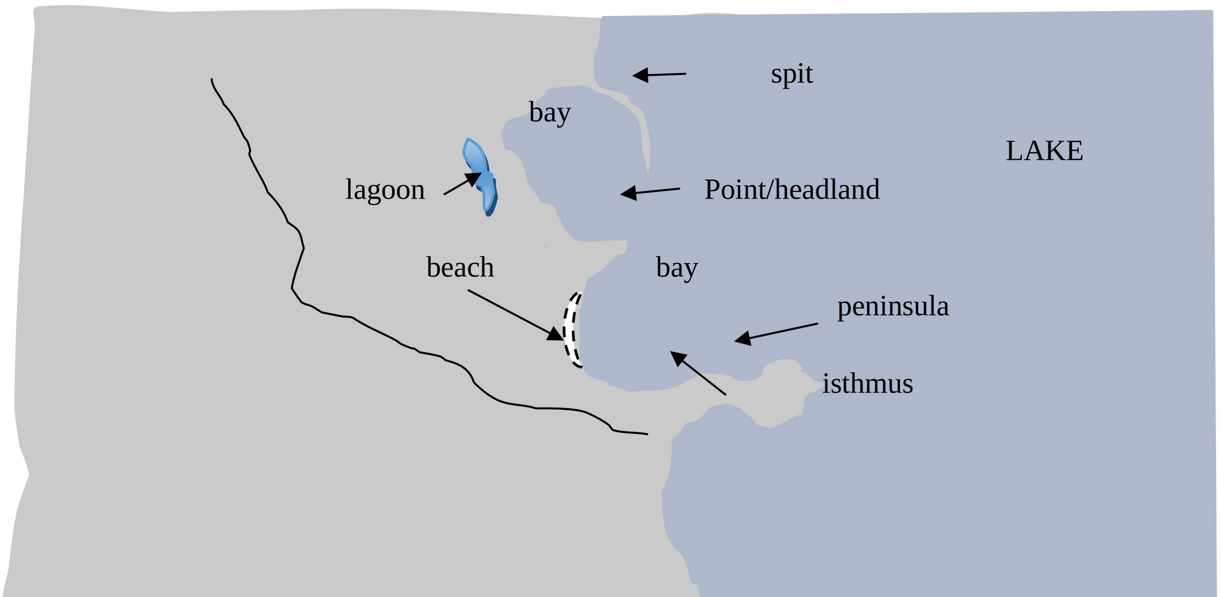
- A beach is a shoreline covered by sand or pebbles.

Lagoon

- A lagoon is a small lake separated from the mother lake. Chia lagoon in Nkhotakota is just one example.

Swamps

- A swamp is a low land that is usually waterlogged. Some of these on Lake Malawi are simply deltas. Examples include Dzadza on Bua river point.



Delta ← shoreline

LANDFORMS IN EAST AND CENTRAL AFRICA

Highlands or mountains

- These include mountains like Kilimanjaro in Tanzania, Kenya in Kenya, Margharita (Ruwenzori) along Uganda and Congo border, and Elgon in Kenya.
- These are Africa's tallest mountains respectively.
- All these are volcanic mountains.
- The others include Ethiopian highlands in Ethiopia.
- Generally, east Africa stands higher than the rest of Africa.
- All these mountains are above 4000m above sea level. Malawi's Mulanje (residual mountain) is 3001m above sea level and is the highest in Central Africa.

Rivers in east and central Africa.

- Rivers such the Nile from Lake Victoria and its tributaries such as Atbara and the Blue Nile both from Ethiopian highlands flows northwards into Mediterranean sea.
- The Congo and its tributaries such as Ubangi, Luapula, Lukuga (the outlet of Lake Tanganyika) and Chambeshi flows to the west into Atlantic Ocean. Zambezi whose source is close to that of that of congo, together with its tributaries such as the Kafue,
- Lwangwa, Mazoe, Hunyani and Shire flow eastwards into Indian Ocean.
- In short river drainage system in this region is either northwards with the Nile as the chief river, or westwards with the Congo as the chief river ,and or eastwads with the Zambezi as the chief river

Industrialization

- This refers to when people start to produce their things in masses through the use of machines and services
- Machines and services play a very impotant role on these

Industry

- This is the process of producing finished goods for economic purposes

Types of industries:

There are three types, namely:

- Primary industry– this involves the production of non finished products such as taw materials e.g. crop cultivation, mining and fishing.
- Secondary or the manufacturing industry – these turn the raw materials into finished products. Examples include includes soap making, sugar making, beer brewing, and biscuit making.
- Tertiary industry –these involves distributing the primary and secondary industry products Examples include transport, marketing, trade, tourism, policing, teaching and financing.

Case study of tourism in Malawi

Factors that promote the Malawi tourism industry

- Physical feature like lakes and mountains mountatins
- Traditional peactices such as dances, food, hospitality
- Decent accommodation
- Good transport systems
- Good communication facilities
- Good services at the hotels

What Malawi is doing to improve its tourism industry

- Building good transport systems
- Building good hotels
- Establishing and protecting wildlife reserves
- Use of well-trained managements personnel in hotels

COMMUNICATION, TRANSPORT AND TRADE

- Communication is the sharing of information and ideas from one place to another

Types of communication

- Verbal communication – e.g. talking directly
- Print communication. e.g newspaters and posters
- Electronic communication e,e, radios, phones and televisions

Importance of communication

- They link people, resources,
- Help people to exchange ideas
- Help to give birth to other industries such as tourism, trade, agriculture etc.
- They reduce working time and expenses.
- It has made the world a global village
- They assist in political stability and socio-economic development of a country.
- They reduce isolation.

Challenges associated with communication

- Vandalism of telecommunication facilities.
- Use of outdated and non functional communication facilities.
- Lack of telecommunication network.
- Lack of knowledge on how to use the telecommunication facilities.
- Language barrier.

TRANSPORT

Types of transport

- land such as road, and rail transport
- water such as ships and boats
- Air tansport such as airplanes

Transport in Malawi

There include land rail and air transport

Most areas in Malawi are accessed by road transport although there a lot of work to nbe done on our roads

Problems of transport in Malawi.

- poor roads due to rains and poor construction pracices
- poor and updated air transport facilities
- non functional rail transport systems
- vandalism to transport system parts
- Malawi

TRADE

Trade is the exchang of goods and servies

Importance of trade

- Help to exchange goods and ideas
- Promote international relations
- Help in exchange of culture
- Facilitate growth of transport systems

Main import commodities of Malawi

- Petroleum products
- Machinery
- Electronics
- Clothes

Malawis main export commodities

- Agricultural products such as tobacco, tea, suger, cotton, peas/ most of these commodities are exported raw, hence, fetching poor prices

Balance of trade in Malawi

Balance of trade refers to the value of commodities which a country exports against the value of commodities a country imports.

- A positive balance of trade is achieved when the value of a country's exports surpasses the value of its imports
- A negative balance of trade occurs when the value of a county's exports is lower than that of its imports

Malawi's balance of trade bas ben mostly negative due to

- Poor weather conditions
- Exporting of unfinished products which fetch poor market prices
- Lack of machinery to boost production of export commodities